

Lesson 7

The Lagrange multiplier test, the principles of testing, and generalized least squares

Announcement. Final exam: December 7th, 2 hours.

1 Testing $R\beta = r$: the Wald idea and the Lagrange multiplier idea

The model and the hypothesis are as before:

$$y = X\beta + u, \quad H_0 : R\beta = r.$$

The Wald idea: is $Rb - r$ far from zero?

Estimate the *unrestricted* model and measure the distance $|Rb - r|$. The distance is measured by the quadratic form of Lesson 5,

$$|Rb - r| \longrightarrow \frac{1}{\sigma^2} (Rb - r)' [R(X'X)^{-1}R']^{-1} (Rb - r) \sim \chi_q^2.$$

The Lagrange multiplier idea: is λ_* far from zero?

Estimate the *restricted* model instead (Lesson 6) and look at the multiplier. If the constraint does not bind, the multiplier is zero, so the hypothesis can be written

$$H_0 : \underbrace{\lambda}_{q \times 1} = \underbrace{0}_{q \times 1},$$

and we measure $|\lambda_* - 0|$ by the quadratic form

$$\lambda_*' [\text{Var}(\lambda_*)]^{-1} \lambda_*.$$

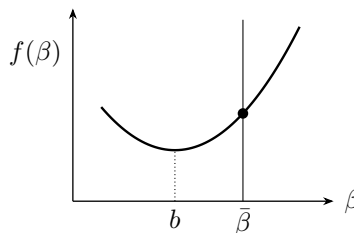


Figure 1: The objective $f(\beta)$ and a constraint (the vertical line). The unconstrained minimum b lies to the left of the line; the constrained solution sits on the line, where the slope of f is not zero. The multiplier measures that slope, the “price” of the constraint: it is zero exactly when the constraint does not bind.

The multiplier and its distribution

From Lesson 6,

$$\begin{aligned}\lambda_* &= [R(X'X)^{-1}R']^{-1}(r - Rb) \\ &= [R(X'X)^{-1}R']^{-1}\left(r - R(\beta + (X'X)^{-1}X'u)\right) \\ &= [R(X'X)^{-1}R']^{-1}(r - R\beta) - [R(X'X)^{-1}R']^{-1}R(X'X)^{-1}X'u.\end{aligned}$$

Under H_0 the first term is zero, so λ_* has mean zero and

$$\begin{aligned}\text{Var}(\lambda_*) &= \mathbb{E}\left[[R(X'X)^{-1}R']^{-1}R(X'X)^{-1}X'uu'X(X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}\right] \\ &= \sigma^2[R(X'X)^{-1}R']^{-1},\end{aligned}$$

using $\mathbb{E}(uu') = \sigma^2 I_n$, after which the middle factors collapse to $R(X'X)^{-1}R'$.

The LM statistic is the Wald statistic

Substituting,

$$\lambda_*' [\text{Var}(\lambda_*)]^{-1} \lambda_* = \frac{1}{\sigma^2} (r - Rb)' [R(X'X)^{-1}R']^{-1} (r - Rb).$$

Since λ_* is $Rb - r$ multiplied by a fixed nonsingular matrix, $|\lambda_*| = |Rb - r|$ in this sense: testing $\lambda_* = 0$ with the *restricted* estimation is the same thing as testing $Rb - r = 0$ with the *unrestricted* estimation.

Replacing σ^2

As before, σ^2 is estimated from the unrestricted residuals,

$$s^2 = \frac{e'e}{n - K} = \frac{u'Mu}{n - K} = \frac{y'My}{n - K},$$

which gives

$$F(q, n - K) = \frac{(r - Rb)' [R(X'X)^{-1}R']^{-1} (r - Rb)/q}{e'e/(n - K)} \sim F(q, n - K).$$

2 Why not the restricted variance estimate?

The restricted estimation gives its own estimate of σ^2 ,

$$s_*^2 = \frac{e'_* e_*}{n} = \frac{\text{RSSR}}{n}.$$

If we intend to use $e'_* e_*/(n - K)$ to replace $e'e/(n - K)$ in the denominator, then the numerator and denominator of the F statistic are no longer independent.

The subset example

Take

$$y = X_1\beta_1 + X_2\beta_2 + u, \quad H_0 : \underbrace{\beta_1}_{k_1 \times 1} = \underbrace{0}_{k_1 \times 1}.$$

From Lesson 6, under H_0 ,

$$F(k_1, n - K) = \frac{u'(M_2 - M)u/k_1}{u'M_2u/(n - K)} \longrightarrow \frac{(\text{RSSR} - \text{USSR})/k_1}{\text{USSR}/(n - K)}.$$

With the restricted residuals in the denominator instead,

$$\frac{u'(M_2 - M)u/k_1}{u'M_2u/(n - K)}.$$

Note that

$$M_2(M_2 - M) = M_2 - M \neq 0,$$

so the numerator and the denominator are not independent, whereas in the previous statistic

$$M(M_2 - M) = M - M = 0$$

(because $MM_2 = M$), which is what makes them independent. The previous one is an F . The one that is not independent has a *beta* distribution: writing $u'M_2u = u'(M_2 - M)u + u'Mu$, the ratio $u'(M_2 - M)u/u'M_2u$ is of the form $A/(A + B)$ with $A/\sigma^2 \sim \chi_{k_1}^2$ and $B/\sigma^2 \sim \chi_{n-K}^2$ independent, which is $\text{Beta}(\frac{k_1}{2}, \frac{n-K}{2})$.

3 Principles of testing

There are three ways to test a restriction:

- ① Estimate the *unrestricted* model only: the **Wald** test.
- ② Estimate the *restricted* model only: the **Lagrange multiplier** (or **score**) test.
- ③ Estimate *both* the restricted and the unrestricted model: the **likelihood ratio** test.

In the linear model with normal errors and a known σ^2 the three coincide, as Section 1 showed for the first two; the RSSR/USSR form of the F statistic uses both estimations.

4 Non-spherical disturbances: generalized least squares

Now we introduce a more general error structure:

$$y = X\beta + u, \quad X \text{ fixed}, \quad u \sim N(0, \sigma^2\Omega), \quad \Omega \neq I_n, \quad \Omega = \begin{bmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{n1} & \cdots & w_{nn} \end{bmatrix}.$$

OLS is still unbiased

$$b = (X'X)^{-1}X'y = \beta + (X'X)^{-1}X'u, \quad \mathbb{E}(b) = \beta,$$

but its variance changes:

$$\text{Var}(b) = \mathbb{E}[(b - \beta)(b - \beta)'] = \sigma^2(X'X)^{-1}X'\Omega X(X'X)^{-1}.$$

Transforming the model

Look for a matrix T such that

$$Tu \stackrel{?}{\sim} N(0, \sigma^2 I) : \quad T \text{Var}(u)T' = \sigma^2 T\Omega T' = \sigma^2 I_n.$$

Does a T with $T\Omega T' = I_n$ exist or not? Refer to the previous lecture (the spectral decomposition, Lesson 5): Ω is symmetric and positive definite, so

$$Q'\Omega Q = \Lambda \implies \Omega = Q\Lambda Q' = PP', \quad P = Q\Lambda^{1/2},$$

where Λ is the diagonal matrix of eigenvalues and Q the matrix of eigenvectors. Then we need

$$TPP'T' = I,$$

and $T = P^{-1}$ works: $P^{-1}PP'(P')^{-1} = I_n$. With $T = P^{-1}$,

$$\Omega = PP' = T^{-1}(T')^{-1}, \quad \Omega^{-1} = T'T.$$

The GLS estimator

Premultiply the model by T :

$$Ty = TX\beta + Tu, \quad Tu \sim N(0, \sigma^2 I_n).$$

Writing $y_* = Ty$, $X_* = TX$, $u_* = Tu$,

$$y_* = X_*\beta + u_*$$

is the standard OLS framework, so

$$b_* = (X_*'X_*)^{-1}X_*'y_* = (X'T'TX)^{-1}X'T'Ty = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y,$$

$$\text{Var}(b_*) = \sigma^2(X_*'X_*)^{-1} = \sigma^2(X'\Omega^{-1}X)^{-1}.$$

It is unbiased:

$$\begin{aligned} b_* &= (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}(X\beta + u) \\ &= (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}X\beta + (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}u, \\ \mathbb{E}(b_*) &= \beta + 0. \end{aligned}$$

OLS against GLS

$$\begin{aligned} b &\sim N(\beta, \sigma^2(X'X)^{-1}X'\Omega X(X'X)^{-1}) && \longrightarrow \text{oblique projection on } L(X), \\ b_* &\sim N(\beta, \sigma^2(X'\Omega^{-1}X)^{-1}) && \longrightarrow \text{orthogonal projection on } L(X_*). \end{aligned}$$

GLS is more efficient if $u \sim N(0, \sigma^2\Omega)$: in the transformed model the Gauss–Markov theorem applies, so b_* is the best linear unbiased estimator, and $\text{Var}(b) - \text{Var}(b_*)$ is positive semi-definite.

5 Two special cases of Ω

Heteroskedasticity: Ω_1 diagonal

$$\Omega \longrightarrow \Omega_1 = \begin{bmatrix} w_{11} & & 0 \\ & \ddots & \\ 0 & & w_{nn} \end{bmatrix} : \quad \text{Var}(u_i) = w_{ii}, \quad \mathbb{E}(u_i u_j) = 0 \ (i \neq j), \quad i = 1, \dots, n.$$

Correlated errors with a common variance: Ω_2

$$\text{or } \Omega_2 = \begin{bmatrix} \bar{w} & w_{12} & \cdots & w_{1n} \\ w_{21} & \bar{w} & & \vdots \\ \vdots & & \ddots & \\ w_{n1} & w_{n2} & \cdots & \bar{w} \end{bmatrix} : \quad \text{Var}(u_i) = \bar{w}, \quad \mathbb{E}(u_i u_j) \neq 0.$$

Transcription notes

- The set opens with a divider page carrying only “Lesson 7”; it is not reproduced. Pages 10 and 11 are blank.
- As in Lessons 3–6 the regressor matrix, written lower case (x) on the page, is typeset X ; B for β is typeset β ; k for the number of regressors is typeset K ; 6^2 is σ^2 . The multiplier, written both λ^* and λ_* , is typeset λ_* as in Lesson 6. The column space, written $\ell(x)$, is typeset $L(X)$ as in Lessons 2–3.
- The GLS estimator is written both b^* and b_* (and x_* , y^* , u^* for the transformed data); typeset b_* , X_* , y_* , u_* throughout. Note that Lesson 6 uses b_* for the *restricted* least-squares estimator; here it is GLS.
- Page 2: the first line “ $|Rb - r| \rightarrow \dots$ ” has a bare $[$ missing before $R(X'X)^{-1}R'$; typeset with brackets. **Corrected:** the first line for λ^* is written $[R(X'x)^{-1}R'](r - Rb)$ without the inverse; the next line has it. The third line is written $\dots - (R(R(X'X)^{-1}R(X'X)^{-1}X'u$ and runs off the page; typeset $[R(X'X)^{-1}R']^{-1}R(X'X)^{-1}X'u$. “Under H_0 the first term is zero” is **added**.
- Page 2, the figure: axes $f(\beta)$ and β , a convex curve and one vertical line to the right of the minimum, with no other labels. The labels b and $\bar{\beta}$ and the caption are **added** (the constraint figure of Lesson 6 has the same layout).

- Page 3: **Corrected:** $\text{Var}(\lambda_*)$ is written without the expectation, $(\dots)X'uu'X(\dots)$; typeset with $\mathbb{E}$. **Corrected:** the quadratic form is written $\frac{1}{\sigma^2}(r - Rb)'(R(X'X)^{-1}R')(r - Rb)$ without the inverse; it is $[R(X'X)^{-1}R']^{-1}$. The annotation “ $|\lambda_*| = |Rb - r|$ ” with an arrow to “restricted” and a bracket to “same thing as $\lambda_* = 0 \rightarrow$ unrestr. estimation” is written out as the paragraph “The LM statistic is the Wald statistic”. In the F numerator the last factor is written $(r - rb)$ and the first lacks a transpose; typeset $(r - Rb)' \dots (r - Rb)$.
- Page 4: “If we intend to to $e'_*e_*/n - k$ replace $e'e$ in denominator” $\rightarrow$ *to use $e'_*e_*/(n - K)$ to replace $e'e/(n - K)$ in the denominator.* **Corrected:** both F statistics on this page are labelled $F(K, n - K)$; with $H_0 : \beta_1 = 0$ the numerator degrees of freedom are k_1 . The dimension labels under H_0 read “ $k_1 \times 1$ $k_1 \times 1$ ”. “previous one is F. Not independent one is β distri.” is read as a *beta distribution*; the reason $(A/(A + B))$ and $MM_2 = M$ are **added**.
- Page 5: the three principles are numbered ①–③ with “Score” written above “Lagrange multiplier” and a large bracket to their right with no label. The sentence on when the three coincide is **added**. “Now we intriduce non-linear” is read as introducing a more general (non-spherical) error covariance, which is what the page goes on to do; “non-linear” is not kept.
- Page 6: $\text{Var}(b)$ is written $\mathbb{E}(b - \beta)(b - \beta)'$; typeset $\mathbb{E}[(b - \beta)(b - \beta)']$. **Corrected:** the decomposition is written “ $\Omega = PP' = Q\Lambda^{1/2}$ ” with Λ labelled “eigen value of diagonal matrix” and Q “eigen vector”; it is $\Omega = Q\Lambda Q' = PP'$ with $P = Q\Lambda^{1/2}$. “Symmetric and positive definite” is **added**. “Refer to previous lecture” is taken to be Lesson 5 (spectral decomposition).
- Page 7: **Corrected:** the transformed model is written $y^* = y^*\beta + u^*$; it is $X_*\beta$.
- Page 8: “ $b \rightarrow$ oblique project. on $\ell(X)$ ” and “ b_* from orthogonal proj. $\ell(X_*)$ ” are kept as written. In the original y -space OLS is the orthogonal projection on $L(X)$ and GLS an oblique one; the page’s statement holds in the transformed space, where $X_*b = TX(X'X)^{-1}X'T^{-1}y_*$ projects onto $L(X_*)$ but is not symmetric. “GLS it’s more efficient if with $u \sim N(0, \sigma^2\Omega)$ ” is kept; the Gauss–Markov reason is **added**. “ $\text{Var}(u_i) = w_{ii} = \mathbb{E}(u_iu_j) = 0$ ” is read as two statements, with $i \neq j$ **added**.
- Page 9: Ω_2 has $\bar{w}$ on the diagonal and $w_{12}, \dots, w_{1n}, w_{21}, w_{n1}, w_{n2}$ off it. The names “heteroskedasticity” and “correlated errors with a common variance” are **added**; the page gives only the matrices and the moments.
- Pages of the original, for tracing back: page 1 the divider; page 2 the exam date, H_0 , the Wald quadratic form, $H_0 : \lambda = 0$, the figure and λ_* ; page 3 $\text{Var}(\lambda_*)$, the LM statistic, s^2 and $F(q, n - K)$; page 4 s_*^2 , the independence remark and the subset example; page 5 the principles of testing, the model with $\sigma^2\Omega$, and b ; page 6 $\text{Var}(b)$ and the construction of T ; page 7 $\Omega^{-1} = T'T$, the transformed model, b_* and $\mathbb{E}(b_*)$; page 8 OLS against GLS and Ω_1 ; page 9 Ω_2 ; pages 10–11 blank.